

Jiaqi Tang

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EDUCATION

University of Illinois Urbana-Champaign

B.S. in Physics, Minor in Computer Science (College of Engineering)

B.S. in Mathematics (College of Liberal Arts and Sciences)

Champaign, IL

Expected May 2027

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- **Cumulative GPA:** 3.89/4.00; James Scholar Candidate; Dean's List, 2024–2026; Illinois Engineering Outstanding Scholarship; Illinois Engineering Achievement Scholarship.
- **Relevant Coursework:** Stochastic Processes, Probability Theory, Numerical Analysis, Differential Equations, Real Analysis, Data Structures, Quantum Information, Computer Systems, Algorithms and Models of Computation.

RESEARCH EXPERIENCE

Computational Social Choice Research (Manuscript in preparation)

Co-author and Undergraduate Researcher | Advisor: Prof. Igor Mineyev

Champaign, IL

May 2026–Present

- Conducted research on dynamic candidate dropout, developing a unified computational framework to compare Plurality, Score, and STAR voting.
- Built a GPU-accelerated simulation calibrated to ANES 2024 data, generating one million Monte Carlo voter draws across seven candidates and 200 voting-elimination rounds; designed four metrics to evaluate welfare and preference fidelity.
- Optimized dropout parameters (α and γ) through adaptive black-box optimization and robustness analysis and found that Score and STAR preserved more aggregate preference information than Plurality under the evaluated dropout settings.

Illinois Risk Lab

Undergraduate Researcher

Champaign, IL

Jan 2026–May 2026

- Conducted research under Prof. Xiaochen Jing on Ethereum MEV sandwich attacks, reconstructing raw on-chain transactions and DEX swap events in Google BigQuery and building a local sandwich-detection pipeline.
- Formulated victim-conditional risk prediction as estimating whether a candidate transaction would be sandwiched from its pre-trade local context; developed XGBoost and Transformer-based models.
- Validated detected transactions against Dune Analytics–curated sandwich tables under strict and relaxed matching criteria; used relative trade-size features, gas-vulnerability proxies, and counterfactual probing to investigate attack triggers and mitigation strategies.

Illinois Mathematics Lab

Undergraduate Researcher

Champaign, IL

Jan 2025–May 2025

- Worked under the supervision of Prof. Daniel Cooney, focusing on stochastic models of cross-scale evolutionary dynamics.
- Derived ODE/PDE systems from Markov-process and mutation dynamics, analyzed continuous-limit stability, and implemented Euler and RK4 solvers to characterize steady-state convergence.

PROFESSIONAL EXPERIENCE

Carl Zeiss IMT (Shanghai) Co., Ltd.

Data Science Intern, IQSCN

Shanghai, China

May 2026–Aug 2026

- **Demand Forecasting & Planning:** Built an end-to-end machine-level demand and revenue forecasting pipeline across 20+ CMM and industrial CT product families using 89,000+ SAP order lines; reconstructed 7,240 complete machines from exploded BOMs, corrected 3–8× quantity distortions, and achieved approximately 8% one-step and 15% three-month cumulative WMAPE through leakage-free walk-forward validation and Low/Base/High scenario forecasting.
- **Working-Capital & Inventory Optimization:** Developed ZEISS Shanghai's first demand-to-working-capital decision model for 6 CONTURA variants and 532 materials, integrating forecasts with multi-level BOM, inventory, lead-time, MOQ, and supplier data; generated ABC-XYZ segmentation and material-level (s , Q) policies that identified model-estimated inventory-release opportunities equivalent to 9% of available stock.
- **CMM Calibration Optimization:** Parsed 39 CAPTUM/SPECTRUM production files containing 48,845 measurement rows and 1,078 valid quality targets; engineered drift, reset, geometry-balance, and angle-interaction features from nine bearing-gap measurements and YX/YZ/XZ settings; modeled initial setup, burn-in, manual adjustment, and final orthogonality as a stochastic sequential optimization problem; combined transition and quality models with Monte Carlo optimization, yielding a model-estimated 75.7% reduction in mean maximum absolute orthogonality error.

Huafu Securities Co., Ltd
Financial Engineering Intern

Shanghai, China
May 2025–Aug 2025

- Engineered financial features (e.g., Herfindahl–Hirschman Index, industry drift, benchmark deviation) for 5,000+ Chinese open-ended funds to quantify portfolio concentration and thematic bias relative to the CSI 300.
- Applied machine learning methods (hierarchical clustering and Gaussian Mixture Models) to group funds and select several funds per cluster for inclusion in an Enhanced Portfolio Optimization strategy.
- Built a Bayesian state-space model with Kalman filtering to infer daily fund holdings from quarterly disclosures, achieving a mean absolute NAV reconstruction error below 0.01 over the evaluation quarter.

University of Illinois Urbana-Champaign – The Grainger College of Engineering
Expert Learning Assistant for PHYS 212 (Electricity & Magnetism) / Peer Mentor

Champaign, IL
Aug 2024–May 2025

- Prepared experimental materials, supported lab design and data analysis, and guided students during laboratory sessions. Identified student challenges and provided feedback to instructors.
- Mentored newly admitted physics students during their transition to UIUC.

SELECTED PROJECTS

Reliable Reinforcement Learning under Imperfect Learned Dynamics

<https://github.com/TanhJK728/world-model-rl>

Jul 2026–Present

- Implemented continuous-control PPO from scratch using a tanh-Gaussian policy, GAE, clipped objectives, KL monitoring, and vectorized rollouts; achieved a mean Pendulum-v1 return of -153 across three seeds, compared with -1199 for a random policy.
- Trained a bootstrapped dynamics-ensemble world model with epistemic-uncertainty estimation; quantified horizon-compounding error under distribution shift.
- Measured the world-model exploitation gap and evaluated uncertainty-aware rollout termination and confidence-weighted policy updates for model-based RL reliability.

HONORS & AWARDS

- Silver Medal, 34th Spanish Physics Olympiad
- American Invitational Mathematics Examination (AIME I), Score: 13/15

Apr 2023
Feb 2023

SKILLS

- **Programming:** Python, MATLAB, SQL
- **Libraries and Tools:** NumPy, pandas, scikit-learn, PyTorch, XGBoost, Google BigQuery, Git, LaTeX
- **Languages:** Mandarin (Native), Spanish (Native), English (Native), Catalan (Advanced)

LEADERSHIP & SERVICE

- Student Senator, University of Illinois Urbana-Champaign Senate

2024–2025